

CIT215: Database Management Systems & DIG212 Web Development II

Programme GPA Calculator and PISS Management System

Due Date: May 8th 2026

Objective:

This project focuses on building a Programme GPA Calculator and PISS Management System. The system will have a web interface connected to a database. It should store and manage data for students, programmes, PISS, courses, grades, GPA, and target GPA calculations. The system will be developed specifically for the CSE and CIT programmes only. Users should be able to calculate their current GPA and determine what grades are needed to reach a target GPA. A super user should be able to add and manage programme structures and PISS records. **This project should be done in groups of 6.**

Project Tasks:

1. Entities and Attributes

Identify the entities needed for the system. Create a total of nine entities. Each entity should have at least six attributes. Each entity must include a primary key. Use foreign keys to show relationships where needed.

2. Normalization

Normalize the database to third Normal Form (3NF). Show the steps from UNF to 1NF, then to 2NF. Remove repeating data. Ensure all attributes depend on the full primary key. Use sample data to show each stage.

3. ERD

Create an ERD using Crow's Foot notation. Show all entities and attributes. Clearly label primary and foreign keys. Show relationships and cardinalities.

4. Sample Data

Add at least three users to each table. Use realistic data. Include data for students, programmes (CSE and CIT only), PISS, courses, grades, GPA, and target GPA requests.

5. Queries

Write SQL queries to retrieve data. Include:

- one-table query
- join query using two or more tables
- GPA calculation query
- target GPA calculation query

6. Stored Procedures

Create at least two stored procedures.

7. Trigger

Create at least one trigger. The trigger should automate a task. For example, it may update the GPA record when a grade is added. Include the SQL code and a short explanation.

8. Web Development

Build a web dashboard using PHP and MySQL.

The system should include:

- signup
- login
- logout
- restricted access

Show the logged-in username on each page.

The GPA features should allow users to:

- enter grades
- calculate GPA
- enter a target GPA
- calculate required grades

All forms should validate input. Protect the system from SQL injection using prepared statements.

Submission Guide:

- **Select the link to the assignment**
- **Select Submission**
- **Submit a zipped file named CIT215_Project_GroupNo containing:**
- **Project Documentation**
 - Entity Relationship Diagram
 - DatabaseName.sql (with dummy data)
 - PHP Files (include all the php files)
 - ReadMe.md with user log in details.
- **Once your file appears in the File submission are click Save Changes.**